Why look beyond pairs of molecules?

Noncovalent interactions (NCIs) decide how molecules pack in crystals, how substrates bind in catalysts, and how drugs fit their targets.

$$E_{\text{int}} = \underbrace{\sum_{\text{pairs}} E_{IJ}^{(2)}}_{\text{two-body}} + \underbrace{E_{\text{int}}^{(3)}}_{\text{three-body}}$$

$E^{(3)}$ is the energy you **miss** if you only add up pairs: mutual polarization (induction) and correlated electron motion (dispersion) of three molecules at once.



Borromean rings: no two are linked, yet the three hold together

The question behind my PhD: which methods capture $E^{(3)}$ correctly, *why*, and how can we predict it cheaply?

3BHET benchmark & decompose → **3BXB** isolate halogen bonds → **Δ-ML** predict iodine → **Impact** crystals & catalysis

3BHET → 3BXB: benchmarking many-body physics

3BHET: diverse trimers



- 10 trimers: hydrogen, halogen, π–π, CH–π, ionic contacts
- CCSD(T)/CBS reference; MP2, DFT, SAPT family tested
- SAPT2+(3): best total energies (7.8% mean error)
- Three-body **dispersion** is where methods disagree most

3BXB: halogen-bonded trimers

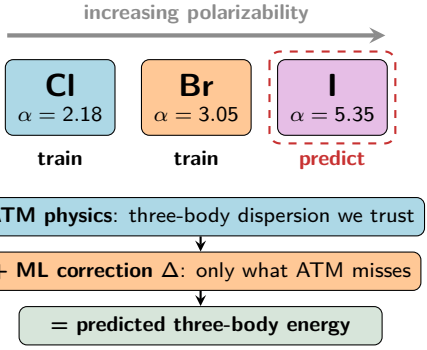


- 214 trimers with Cl, Br, I donors, built from an XB dimer set
- Directional σ-hole bonding: one interaction type, isolated
- Three-body effects grow with halogen polarizability
- Iodine: the most important *and* most expensive case

New problem: iodine needs large basis sets, relativistic treatment and costly CCSD(T). **Can we predict it without computing it?**

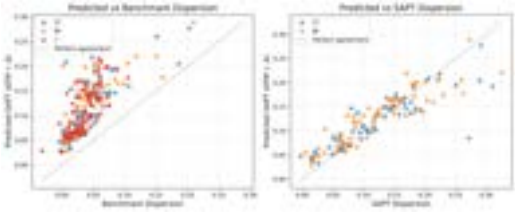
Ochieng & Patkowski, *PCCP* **2023**, 25, 28621. Kříž & Řezáč, *PCCP* **2022**, 24, 14794. Ochieng & Patkowski, *arXiv* **2025**, 2507.17619.

Physics-informed Δ-learning: predicting iodine



$$E^{(3)}_{\mathrm{ML}} = E^{(3)}_{\mathrm{ATM}} + \Delta_{\mathrm{ML}}$$

- Physical features: distances, ATM angles, α, C_6
- Models chosen by **Cl↔Br transfer**, not training fit
- Iodine never seen in training: true extrapolation

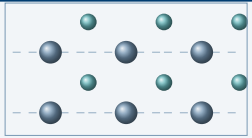


Predicted vs. benchmark and SAPT(DFT) three-body energies

for the unseen iodine systems

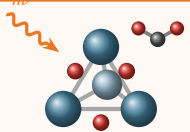
The bigger picture: from trimers to crystals and catalysts

Crystal structure



Every molecule in a crystal feels many neighbors at once. Three-body dispersion and induction shift lattice energies and can change which **polymorph** is predicted to be most stable.

Catalysis



Before CO_2 is activated on a metal cluster it sits in a weakly bound **pre-reactive complex**: cluster, substrate and support form a many-body system. Halogen-bond (σ-hole) catalysts use the same physics.

Prediction at scale

costly reference (CCSD(T))

↓

cheap physics baseline

↓

+ learned Δ correction

The Δ-learning recipe transfers: a DFT or TDDFT baseline plus a learned correction can reach heavier elements and larger clusters (3d→4d→5d metals) at a fraction of the cost.

Accurate many-body interactions, explained by decomposition and scaled with physics-informed ML, are a foundation for designing crystals and catalysts.